Employee Attrition Analysis — Summary Report

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Dataset: IBM HR Analytics | 1,470 Employee Records

Purpose of This Report

This report summarizes findings from a machine learning analysis conducted on IBM's HR employee dataset. The goal was to understand why employees leave the company and to build a system that can identify employees who are at risk of leaving — before they actually do. All findings are based on data from 1,470 employees.

What the Data Tells Us

Out of 1,470 employees in the dataset, 237 employees left the company — an overall attrition rate of 16.1%. This means roughly 1 in 6 employees resigned. The data is skewed — most employees stayed — which makes early identification of at-risk employees even more important.

Key patterns observed in the data:

- **Sales Representatives have the highest exit rate** — 39.76% of them left. That is nearly 4 out of every 10 employees in this role, which is significantly higher than any other job role in the company.
- **Overtime is the strongest predictor of attrition.** Employees who were regularly asked to work beyond regular hours were far more likely to resign. This factor came up as the most important signal in our prediction model.
- **Lower-paid employees leave more often.** Employees who left were earning about 27% less per month on average compared to employees who stayed (median: Rs. 4,969 vs Rs. 6,833). Salary is a real factor — but not the only one.
- **The first two years are the most critical.** Attrition is highest in an employee's first one to two years at the company. Employees who complete five or more years are significantly less likely to leave.
- **Work-life balance matters, but does not tell the whole story.** Employees who rated their work-life balance as poor (1 out of 4) showed a 31% attrition rate. However, even employees with good ratings left — suggesting that salary, overtime, and job satisfaction play equally important roles.

How the Prediction Model Works

We trained and tested three different machine learning models on this data. The best-performing model — Logistic Regression — was able to correctly identify approximately 62% of employees who went on to leave the company. In practical terms, out of every 10 employees who were going to resign, the model would flag about 6 of them in advance, giving HR enough time to intervene.

Recommendations for HR

1. Address overtime as a priority.

Excessive overtime is the single strongest reason employees consider leaving. We recommend reviewing overtime practices — especially in the Sales and R&D; teams — and ensuring that extra hours are fairly compensated. Even a clear and consistent policy around overtime can make a significant difference in how employees feel about the company.

2. Create a structured check-in program for new employees.

Since attrition is highest in the first two years, we recommend formal one-on-one check-ins at the 90-day and one-year marks for every new joiner. Catching dissatisfaction early — before an employee starts looking elsewhere — is far more cost-effective than replacing them after they have already decided to leave.

Important Note

This model is a screening tool, not a final decision. It helps HR identify employees who may be at risk based on patterns in data — but it should always be used alongside direct conversations with managers and the employees themselves. Data can point to a problem; people need to solve it.